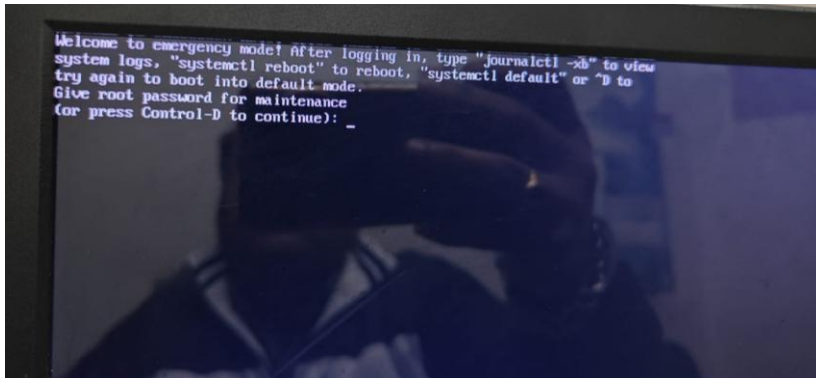


TROUBLESHOOTING THE EMERGENCY MODE ISSUES

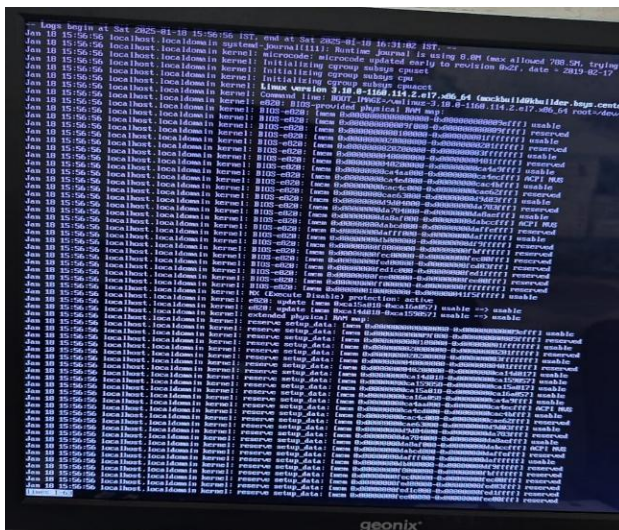
After starting the Linux server, the system went into emergency mode like this:



After logging, to find the what is reason for these issues follow the below-mentioned steps to find out.

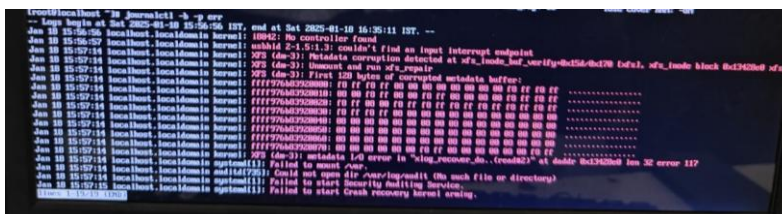
STEP 1: After running the below-mentioned command, the output shows like this

- journalctl -xb



STEP 2: To find out the exact error run this command

- journalctl -b -p err



Several critical issues are evident in the logs, such as XFS filesystem errors and Service failures.

STEP 3: To find out system failed issues, run this command to show only error-level messages from the current boot

- systemctl list-units --failed

```
[root@localhost ~]# systemctl list-units -failed
systemctl: invalid option -- 'e'
[root@localhost ~]# systemctl list-units --failed
UNIT                                LOAD    ACTIVE SUB    DESCRIPTION
■ var.mount                         loaded failed failed /var
■ auditd.service                   loaded failed failed Security Auditing Service
■ kdump.service                    loaded failed failed Crash recovery kernel arming

LOAD    = Reflects whether the unit definition was properly loaded.
ACTIVE  = The high-level unit activation state, i.e. generalization of SUB.
SUB      = The low-level unit activation state, values depend on unit type.

3 loaded units listed. Pass --all to see loaded but inactive units, too.
To show all installed unit files use 'systemctl list-unit-files'.
[root@localhost ~]#
```

The /var directory isn't mounted. So, that is the issue for the system going to emergency mode because that /var directory is mentioned in the fstab file.

STEP 4: Check fstab file entries

- cat /etc/fstab

```
/dev/mapper/centos-home on /home type xfs (rw,relatime,fmask=0077,dmask=0077,codepage=437,iocharset=ascii,sh
[root@localhost ~]# cat /etc/fstab
#
# /etc/fstab
# Created by anaconda on Mon Apr  8 11:53:52 2024
#
# Accessible filesystems, by reference, are maintained under '/dev/disk'
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info
#
/dev/mapper/centos-root /                    xfs     defaults        0 0
UUID=9e17889c-cb2c-4886-8977-82dc3b6c65e4 /boot              xfs     defaults        0 0
/dev/mapper/centos-home /home             vfat    umask=0077,shortname=winnt 0 0
/dev/mapper/centos-var /var              xfs     defaults        0 0
/dev/mapper/centos-swap swap              xfs     defaults        0 0
[root@localhost ~]#
```

In fstab the /var directory is mentioned.

STEP 5: Check the disk space

- df -hT

```
[root@localhost ~]# df -hT
Filesystem                                Type      Size  Used Avail Use% Mounted on
devtmpfs                                 devtmpfs  7.7G   0    7.7G  0% /dev
tmpfs                                     tmpfs     7.8G   0    7.8G  0% /dev/shm
tmpfs                                     tmpfs     7.8G  8.7M   7.7G  1% /run
tmpfs                                     tmpfs     7.8G   0    7.8G  0% /sys/fs/cgroup
/dev/mapper/centos-root                  xfs       20G   2.8G   17G  15% /
/dev/sda2                                xfs       20G  190M   20G   1% /boot
/dev/sda1                                vfat      20G   12M   20G   1% /boot/efi
/dev/mapper/centos-home                  xfs       20G   33M   20G   1% /home
```

- lsblk


```

root@localhost ~# lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINT
sda          8:0    0 931.5G  0 disk
├─sda1       8:1    0  19.5G  0 part /boot/efi
├─sda2       8:2    0  19.5G  0 part /boot
├─sda3       8:3    0 892.1G  0 part
│   └─centos-root 253:0    0  19.5G  0 lvm /
│       └─centos-swap 253:1    0  19.5G  0 lvm [SWAP]
│           └─centos-home 253:2    0  19.5G  0 lvm /home
└─centos-var 253:3    0 833.6G  0 lvm
root@localhost ~#

```

That /var directory isn't mounted

STEP 6: Let's try to mount that /var directory manually

- mount /var

```

root@localhost ~# mount /var
[ 3686.237715] XFS (dm-3): Metadata corruption detected at xfs_inode_buf_verify+0x154/0x170 [xfs], xfs_inode block 0x13420e0 xfs_inode_buf_verify
[ 3686.239553] XFS (dm-3): Unmount and run xfs_repair
[ 3686.241353] XFS (dm-3): First 128 bytes of corrupted metadata buffer:
[ 3686.244982] ffff76b83390000: f8 ff f8 ff 00 00 00 00 00 00 00 00 00 00 f8 ff f8 ff .....
[ 3686.246082] ffff76b83390010: 00 00 f8 ff 00 00 f8 ff 00 00 00 00 00 00 f8 ff f8 ff .....
[ 3686.246882] ffff76b83390020: f8 ff 00 00 f8 ff f8 ff f8 ff f8 ff f8 ff f8 ff f8 ff .....
[ 3686.248616] ffff76b83390030: f8 ff f8 ff 00 00 00 00 00 00 00 00 00 00 00 00 .....
[ 3686.250331] ffff76b83390040: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
[ 3686.252020] ffff76b83390050: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
[ 3686.253608] ffff76b83390060: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
[ 3686.255320] ffff76b83390070: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
[ 3686.256941] XFS (dm-3): metadata L0 error in "xlog_recover_do...(read#2)" at dadr 0x13420e0 len 32 error 117
mount: mount /dev/mapper/centos-var on /var failed: Structure needs cleaning
root@localhost ~#

```

Manual mounting isn't working either, so the filesystem has to be repaired.

STEP 7: Run xfs_repair with the force option to solve these issues because in this server they are using the xfs filesystem.

- xfs_repair -L -v /dev/mapper/centos-var

If that doesn't work, try with maximum verbosity and force options

- xfs_repair -L -v -V /dev/mapper/centos-var

```

agi unlinked bucket 20 is 19737309 in ag 0 (inode=19737309)
agi unlinked bucket 61 is 19536317 in ag 8 (inode=19536317)
agi free 2763, counted 51679436 in ag 1
sb-fsblocks: 26762279, counted 26762286
- found root inode chunk
Phase 3 - for each fs...
- scan and clear agi unlinked lists...
- process known inodes and perform inode discovery...
- agno = 0
- agno = 1
imp claim a free inode 537062380 is in use, correcting imp and clearing imp
cleared inode 537062380
imp claim a free inode 537062387 is in use, correcting imp and clearing imp
cleared inode 537062387
- agno = 2
- agno = 3
- process newly discovered inodes...
- check for duplicate blocks...
- setting up duplicate extent list...
- check for inodes claiming duplicate blocks...
- agno = 0
- agno = 1
- agno = 2
- agno = 3
Phase 5 - rebuild AG headers and trees...
- agno = 0
- agno = 1
- agno = 2
- agno = 3
- reset superblock...
- check inode connectivity...
- reconstruct contents of realtime hitmap and summary inodes
- traversing filesystem...
- agno = 0
- agno = 1
- agno = 2
- agno = 3
- traversal finished...
disconnected inode 19536317, moving to lost-found...
disconnected inode 19737309, moving to lost-found...
disconnected inode 19737309, moving to lost-found...
disconnected inode 19737309, moving to lost-found...
disconnected inode 19737309, moving to lost-found...
disconnected inode 19737309, moving to lost-found...
Phase 7 - verify and correct link counts...
Rebuild metadata sum (115179000) is ahead of log (110).
Format log to cyclic 1151.
XFS_REPAIR Summary Sat Jan 18 17:49:30 2020
Phase Start End Duration
Phase 1: 01/18 17:49:00 01/18 17:49:09 1 seconds
Phase 2: 01/18 17:49:13 01/18 17:49:13 4 seconds
Phase 4: 01/18 17:49:13 01/18 17:49:24 11 seconds
Phase 5: 01/18 17:49:24 01/18 17:49:24 7 seconds
Phase 6: 01/18 17:49:31 01/18 17:49:31 7 seconds
Total run time: 23 seconds
done
root@localhost ~#

```

After running the 1st command xfs_repair -L -v /dev/mapper/centos-var the issues are solved.

STEP 8: After the repair is completed successfully, try to mount it

- `mount /dev/mapper/centos-var /var`

```

root@localhost ~]# mount /dev/mapper/centos-var /var
mount: var: NO such file or directory
root@localhost ~]# df -hT
Filesystem                Type      Size  Used Avail Use% Mounted on
devtmpfs                  devtmpfs  7.7G   0    7.7G   0% /dev
tmpfs                     tmpfs     7.8G   0    7.8G   0% /dev/shm
tmpfs                     tmpfs     7.8G  8.7M   7.7G   1% /run
tmpfs                     tmpfs     7.8G   0    7.8G   0% /sys/fs/cgroup
/dev/mapper/centos-root    xfs       20G    2.8G   17G   15% /
/dev/sda2                 xfs       20G   190M   20G    1% /boot
/dev/sda1                 vfat      20G    12M   20G    1% /boot/efi
/dev/mapper/centos-home    xfs       20G    33M   20G    1% /home
/dev/mapper/centos-var     xfs       834G   41G   793G    5% /var
root@localhost ~]#

```

Now the manual mounting is working ensure it is mounted in the disk table.

- `df -hT`
- `lsblk`

```

root@localhost ~]# df -hT
Filesystem                Type      Size  Used Avail Use% Mounted on
devtmpfs                  devtmpfs  7.7G   0    7.7G   0% /dev
tmpfs                     tmpfs     7.8G   0    7.8G   0% /dev/shm
tmpfs                     tmpfs     7.8G  8.7M   7.7G   1% /run
tmpfs                     tmpfs     7.8G   0    7.8G   0% /sys/fs/cgroup
/dev/mapper/centos-root    xfs       20G    2.8G   17G   15% /
/dev/sda2                 xfs       20G   190M   20G    1% /boot
/dev/sda1                 vfat      20G    12M   20G    1% /boot/efi
/dev/mapper/centos-home    xfs       20G    33M   20G    1% /home
/dev/mapper/centos-var     xfs       834G   41G   793G    5% /var
root@localhost ~]# lsblk
NAME            MAJ:MIN RM  SIZE RO TYPE MOUNTPOINT
sda              8:0    0 931.5G  0 disk
├─sda1           8:1    0  19.5G  0 part /boot/efi
├─sda2           8:2    0  19.5G  0 part /boot
├─sda3           8:3    0  892.1G  0 part
│   ├─centos-root 253:0    0  19.5G  0 lvm /
│   ├─centos-swap 253:1    0  19.5G  0 lvm [SWAP]
│   ├─centos-home 253:2    0  19.5G  0 lvm /home
│   └─centos-var  253:3    0  833.6G  0 lvm /var
root@localhost ~]#

```

STEP 9: Reboot the system and ensure now the issue is solved or not.

- Reboot

```
CentOS Linux 7 (Core)
Kernel 3.10.0-1160.114.2.el7.x86_64 on an x86_64

localhost login: root
Password:
Last login: Sun Jan 12 16:00:26 from 103.214.61.152

#####
Welcome to Naudabao Telephony System
#####

[root@localhost ~]#
```

Now the issue is solved.